PAPER_384 — Sagittarius A* Full Resonance + Compressed Term Decomposition

Source: grok_share_11254865.txt, lines ~2960–2990
Section: Sagittarius A* resonance and compressed MUGE computation with per-term values
Session: 104 (Complete Re-Analysis — full per-term decomposition for Sag A* undiscovered)
CP4 Class: `SagAStarFullResonanceTermDecompositionCalculator` (CP4 #35)

Abstract

This paper presents a UQFF analysis of Sagittarius A* Full Resonance + Compressed Term Decomposition, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

PAPER_371 (resonance MUGE framework) and PAPER_372 (compressed MUGE framework) described the equations and final results for multiple systems including Sagittarius A. *PAPER_379 compared compressed vs resonance final totals. However, the* ***individual per-term values for Sag A*** were never explicitly tabulated in any paper.

This paper provides the **first per-term decomposition** for the Sagittarius A* SMBH under both MUGE models, revealing that Sag A* exhibits different dominant terms than SGR1745 (PAPER_382), and demonstrating a consistent **fluid-dominance law** across both compact object and SMBH scales.

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2. Sagittarius A* System Parameters

Parameter	Symbol	Value	Units
Mass	M	8.155e36	kg
Radius	r	1×10^{12}	m
Magnetic field	B	1×10^{-5}	T
Critical B-field	B_crit	1×10^{-4}	T
Age	t	3.786e14	s
Redshift	z	0.0009	—
V_sys	3.552e45	m^3	
v_exp	5×10^6	m/s	
f_fluid	3.465e-8	Hz	
Current I	1×10^{23}	A	
Area A	2.813e30	m^2	

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3. Resonance MUGE — Per-Term Decomposition

Term 1: aDPM

$$F_{DPM} = I \cdot A \cdot (\omega_1 - \omega_2) = 10^{23} \times 2.813 \times 10^{30} \times 10^9 = 2.813 \times 10^{62}$$

$$a_{DPM} = F_{DPM} \cdot f_{DPM} \cdot E_{vac,neb} \cdot c \cdot V_{sys}$$

$$\boxed{a_{DPM}^{\text{SgrA}*} = 1.001 \times 10^{-10} \text{ m/s}^2}$$

Term 2: aTHz

$$a_{THz} = f_{THz} \cdot \frac{E_{vac,neb} \cdot v_{exp} \cdot a_{DPM}}{E_{vac,ISM} \cdot c}$$

With $v_{exp} = 5 \times 10^6$ m/s for Sag A*:

$$\boxed{a_{THz}^{\text{SgrA}*} = 1.001 \times 10^{-2} \text{ m/s}^2}$$

Term 3: Fluid Frequency Coupling (afluid_freq) — DOMINANT

$$a_{fluid\_freq} = f_{fluid} \cdot \frac{E_{vac,neb} \cdot V_{sys}}{E_{vac,ISM} \cdot c}$$

With Sag A*: $f_{fluid} = 3.465 \times 10^{-8}$ Hz, $V_{sys} = 3.552 \times 10^{45}$ m^3:

$$\boxed{a_{fluid\_freq}^{\text{SgrA}*} = 4.105 \times 10^{29} \text{ m/s}^2 \quad \textbf{(DOMINANT)}}$$

Remaining resonance terms for Sag A*

Term	Value (m/s^2)	Note
avac_diff	$\sim 10^{-12}$	small — low Δ_Evac
asuper_freq	$\sim 10^{-5}$	B-field much weaker than magnetar
aaether_res	$\sim 10^{-28}$	sub-dominant
Ug4i	≈ 0	ancient system (t=3.786e14 s)
aquantum_freq	$\sim 10^{-60}$	Hubble quantum floor
aAether_freq	$\sim 10^{-74}$	minimum
Osc_term	0	steady state
aexp_freq	$\sim 10^{-47}$	cosmological
fTRZ	0.1	parametric coupling constant

Resonance MUGE total: $\approx a_{fluid\_freq} = 4.105 \times 10^{29}$ m/s^2 (fluid-dominated)

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4. Compressed MUGE — Per-Term Decomposition

Term 1+2: Newtonian + SC adjustment

$$g_{\text{base}} = \frac{GM}{r^2} = \frac{6.674 \times 10^{-11} \times 8.155 \times 10^{36}}{(10^{12})^2}$$

$$g_{\text{base}} = 5.443 \times 10^2 \text{ m/s}^2$$

Superconducting adjustment (B/B_crit = $10^{-5}/10^{-4}$ = 0.1):

$$g_{\text{SC}} = 5.443 \times 10^2 \times 0.9 = 4.899 \times 10^2 \text{ m/s}^2$$

Term 3: Fluid coupling

$$g_{\text{fluid}}^{\text{SgrA*}} = \rho_{\text{fluid}} \cdot V_{sys} \cdot g_{\text{local}}$$

$$\boxed{g_{\text{fluid}}^{\text{SgrA*}} = 3.552 \times 10^{20} \text{ m/s}^2}$$

Term 4: Dark Matter Perturbation (DOMINANT in compressed model)

$$g_{\text{pert}} = (M + M_{DM}) \left(\frac{\delta\rho}{\rho} + \frac{3GM}{r^3} \right)$$

With $3GM/r^3 = 3 \times 6.674 \times 10^{-11} \times 8.155 \times 10^{36}/(10^{12})^3 = 1.63 \times 10^{-15}$:

Note: Sag A* has large M but much larger r compared to magnetar $\rightarrow$ lower $3GM/r^3$:

$$g_{\text{pert}}^{\text{SgrA*}} \approx (M + M_{DM}) \times (0.1 + 1.63 \times 10^{-15}) \approx (M + M_{DM}) \times 0.1$$

$$\boxed{g_{\text{pert}}^{\text{SgrA*}} = 2.966 \times 10^{34} \text{ m/s}^2}$$

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5. Complete Comparison Table: Sag A* Both Models

Resonance MUGE

Term	Value (m/s^2)	Orders above aDPM
afluid_freq	**4.105e29**	+39
aTHz	1.001e-2	+32
aDPM	1.001e-10	—
All others	negligible	—

Total resonance: ≈ 4.105e29 m/s^2

Compressed MUGE

Term	Value (m/s^2)	Notes
Perturbation	**2.966e34**	DOMINANT
Fluid	3.552e20	sub-dominant by 14 orders
Newtonian SC	4.899e2	base

Total compressed: ≈ 2.966e34 m/s^2

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6. Cross-Model Comparison: Sag A* vs SGR1745

Property	SGR1745 (Magnetar)	Sag A* (SMBH)
Resonance dominant term	afluid=1.773e-9	afluid=4.105e29
Resonance dominant mechanism	vacuum×volume coupling	vacuum×volume coupling
Compressed dominant term	perturbation=1.782e39	perturbation=2.966e34
Compressed/Resonance ratio	$\sim 10^{48}$	$\sim 10^{5}$
Resonance total (m/s^2)	1.773e-9	4.105e29
Fluid term ratio Sag A*/SGR1745	—	×2.3e38

Fluid Universality Principle: The dominant resonance term in both a compact magnetar ($r = 10^4$ m) and a supermassive black hole ($r = 10^{12}$ m) is a_{fluid_freq}, but the values differ by **38 orders of magnitude**, scaling with $f_{fluid} \cdot V_{sys}$.

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7. Physical Interpretation: Why Sag A* Fluid Term Dominates So Strongly

The fluid frequency coupling scales as:

$$a_{fluid\_freq} \propto f_{fluid} \cdot V_{sys}$$

For Sag A*: - $V_{sys} = 3.552 \times 10^{45}$ m^3 (sphere of radius 1 pc around SMBH) - $f_{fluid} = 3.465 \times 10^{-8}$ Hz (frequency of fluid oscillation in SMBH accretion disk)

The product $f_{fluid} \cdot V_{sys} = 3.465 \times 10^{-8} \times 3.552 \times 10^{45} = 1.231 \times 10^{38}$ m^3/s

Versus SGR1745: $f_{fluid} \cdot V_{sys} = 1.269 \times 10^{-14} \times 4.189 \times 10^{12} = 5.315 \times 10^{-2}$ m^3/s

Ratio: $1.231 \times 10^{38} / 5.315 \times 10^{-2} \approx 2.3 \times 10^{39}$ — explaining the 39-order dominance difference between the two fluid terms.

The SMBH has an astronomically larger vacuum energy coupling volume, making its resonance fluid term the largest of any system in the canonical 7-system registry.

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8. References Within Codebase

- PAPER_371: MUGE 12-Term Resonance — Sag A* final result (fluid dominant)
- PAPER_372: Compressed UQFF — Sag A* compressed result
- PAPER_379: Dual-Model Comparison — totals side-by-side (SGR1745 vs others)
- PAPER_381: SGR1745 compressed decomposition — comparison baseline
- PAPER_382: SGR1745 resonance decomposition — comparison baseline
- `MUGESuperconductive12TermResonanceCalculator` (CP4 #14): Sag A* via `sagA_dataset`

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